Storm Time Seasonal Variation of TEC in the Southern Hemisphere Mid-Latitude Regions Using Signals from GPS Satellite

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Abstract: Total Electron Content (TEC) of the ionosphere over the Australian regions has been derived from measurements obtained at GPS ground stations of the International GPS Service (IGS). The derived TEC values at mid- and low-latitudes during the severe magnetic storms of 6 April, 15 July, and 6 November 2000 (equinox, winter, and summer, respectively, for the southern hemisphere), show large deviations from the quiet day behaviour.). Both an enhancement and a depletion of TEC relative to quiet day are observed during the aforementioned storms and are called positive and negative storm effects, respectively. The observations, in general, indicate an enhanced plasma loss over 24 hours after the storm began, which is probably due to the equatorward expansion of storm induced composition changes. Both zonal electric fields and meridional winds are assumed to contribute to the ionospheric negative storm effect propagating from higher to lower latitudes. Comparisons of GPS TEC with f_0F_2 data from a wide area network of ionosondes in the Australian region show good agreement with a correlation coefficient greater than 0.83.

Keywords: Ionosphere, Magnetic storm, TEC, and GPS